Email: qulip.louis.0126@gmail.com
Mobile: +886-906024704
GitHub: LouisChang0126
Website: louischang0126.github.io

Louis(Chia-Jui) Chang

Education

National Yang Ming Chiao Tung University Sep. 2026 - Jun. 2028 (expected)
M.S. in Data Science and Engineering, College of Computer Science

National Yang Ming Chiao Tung University Sep. 2022 - Jun. 2026
B.S. in Computer Science: GPA: 4.02/4.3 (overall)

Selected Coursework
Robotics & Vision: Physical AI *(Grad)*, Robotic Navigation and Exploration *(Grad)*, Selected Topics in Visual Recognition using DL *(Grad)*, AI Capstone, Embedded Systems Capstone.
AI & ML: Artificial Intelligence *(Grad)*, Machine Learning, Natural Language Processing.
CS Foundations: Data Structures & OOP, Algorithms, Operating Systems, Computer Networks.
Math: Linear Algebra, Probability, Discrete Mathematics, Applied Statistical Methods.
Security & Systems: Cryptography Engineering *(Grad)*, Computer System Administration.

Technical Skills

Languages: Python, C/C++, JavaScript, HTML, PHP
ML / DL: PyTorch, TensorFlow, ONNX, OpenCV, scikit-learn, Hugging Face, Whisper (STT)
Computer Vision: YOLO, D-FINE, SAM3, CLIP, Diffusion Models, Model Quantization, Semi-supervised Learning
Robotics: ROS2, PX4, Gazebo, Universal Manipulation Interface, Stretch3
LLM / GenAI: Vibe Coding, Fine-tuning LLMs, Model Routing, LLM API Integration, MCP Design
Cloud / Backend: GCP (Cloud Functions, Firestore, Cloud Scheduler), LINE Bot SDK
Tools: Git, Linux, Arduino, 3D Modeling, SMT Soldering

Award

2024 Meichu Hackathon - LINE Taiwan 3rd Place Link **Aug. 2024**

- **Led** 5-member team in 24-hour competition; proposed a building-classification model to replace GPS and resolve inaccurate positioning indoors.
- **Built** a LINE Bot frontend and integrated Gemini + GCP on the backend to identify nearby buildings from user photos and guide them to the destination.

2025 Singapore Amazing Flying Machine Competition (SAFMC) - D2 Category 4th Place **Mar. 2025**

- Joined 18-member team to develop a system enabling 4 drones to perform autonomous and synchronized pick-and-place operations.
- **Software:** implemented the final state machine, integrated Gazebo simulation, developed the mediator module using ROS2, and tuned PX4 parameters.
- **Hardware:** designed 3D-printed components and performed SMT soldering.

NYCU CS Project Competition - High Distinction Award GitHub1 2 **Jun. 2025**

- Organized and open-sourced SAFMC results; competed under the title Multi-Drones Automatically Pick & Place.

2025 Make NTU - AUO 3rd Place **May. 2025**

- **Developed** an intelligent parking system maximizing car storage density, combining STT, LLM, and TTS on the server side with an automated lift-and-transport robot car.
- Responsible for project conceptualization, team leadership, mechanical design, and programming the vehicle's movement.

Experience

Garmin AOEM R&D Internship **Jul. 2025 - Feb. 2026**

- **Road-Hazard Segmentation** (in-vehicle driver safety): **automated** the annotation pipeline with SAM3, **synthesized** hazardous-scenario training samples via diffusion models, and accelerated the training workflow by 135%.
- **SE-Asia Road Object Detection: fine-tuned** YOLOX via semi-supervised learning to transfer a Taiwan-trained model to unlabeled regional imagery; additionally trained D-FINE and applied model quantization to optimize accuracy and latency for edge-AI deployment.

Uancare Spring Semester Internship **Feb. 2025 - Jun. 2025**

- **Oral Movement Detection: built** an in-app model to help seniors monitor oral muscle health; processed 10+ hours of video into labeled image datasets and trained multiple YOLO variants, achieving **>0.85 recall across all classes**.

- **Diadochokinetic (DDK) Audio Analysis: reproduced** academic CNN-based DDK methods and trained a custom model with performance comparable to traditional DDK approaches.

Human-centered Intelligent System Lab (HCIS Lab) **Jun. 2024 - Present**

Contributing to the lab across my undergraduate years and continuing into my Master's studies:

- **Led** 7-member team researching physical AI on a mobile manipulator in Fall 2024.
- **Extended** the lab's work into autonomous multi-drone systems in Spring 2025, developing synchronized pick-and-place across 4 UAVs with ROS2 + PX4 + Gazebo.
- **Built** a real-to-sim robot arm pipeline: collected real-world demonstrations via Universal Manipulation Interface, transferred them into the RoboCasa simulator, and trained manipulation policies with Diffusion Policy; the pipeline was adopted as the curriculum of FunAI 2025.
- **Customized** the Stretch3 gripper and a handheld gripper with multi-modality capabilities, drawing inspiration from CLAMP and Robot Utility Models.

Intro. to AI Teaching Assistant GitHub **Jan. 2025 - Jun. 2025**

- **Designed** a supervised-learning programming lab comparing CNN vs. decision-tree classifiers for 100+ undergraduates in Prof. Yi-Ting Chen & Prof. Yu-Chun Yen's Intro. to AI class.

Publications

Beyond Memorization: A Multi-Modal Ordinal Regression Benchmark to Expose Popularity Bias in Vision-Language Models

Li-Zhong Szu-Tu, Ting-Lin Wu*, **Chia-Jui Chang**, He Syu, Yu-Lun Liu*

Paper | GitHub | HuggingFace | Project Page

YearGuessr introduces a global, multi-century dataset of building ages. We propose YearCLIP and benchmark over 30 other models, ultimately highlighting the popularity bias of VLMs in this task.
An earlier version, *"YearGuessr: A Worldwide Building Age Benchmark with 55k Wikipedia Images and Ordinal Targets"* was accepted as an **Oral Presentation at ICCVW'25 DataCV**. The advanced version was submitted to NeurIPS DB Track, receiving scores of 4, 5, 5, 5 (borderline accept * 1 + accept * 3) after the rebuttal, narrowly missing acceptance despite strong reviews.

Projects

Built Year Prediction — Final Project, Intro. to AI GitHub **Jun. 2024**

- **Trained** a ResNet-50 via transfer learning to predict building construction year from images; labeled the dataset with Gemini.
- **Applied** Grad-CAM to visualize regions the CNN attended to when predicting year; awarded **Best Presentation** in the final project competition.

Church Service Scheduling LINE Bot — Personal Project GitHub **Jul. 2023 - Present**

- **Built** an end-to-end scheduling system serving **30+ members for 2+ years** across multi-service church operations: LINE Bot for shift swap & service reminders, plus a web dashboard for browsing and editing schedules; backed by GCP Cloud Functions and Firestore.
- **Integrated** an LLM agent that auto-generates and modifies shift assignments from CSV personnel data under custom rules (e.g., avoiding consecutive assignments, capping weekly service frequency), **reducing manual scheduling time by 85%**.

Arduino Macro Keyboard — Personal Project **Apr. 2023**

- **Designed** a customizable macro keyboard on Arduino Pro Micro with user-defined macros via firmware; taught 61 students across 2 club courses as Teaching Officer of NYCU Makereal Club.

Extracurricular Activities

NYCU Koinonia Club *President* **Sep. 2022 - Jun. 2026**

- **Organized** recurring events including English Table, Club Day, Xmas Party, and Poetry Performance.

NYCU Makereal Club *Teaching Officer* **Sep. 2022 - Jan. 2024**

- **Developed** hands-on maker projects (Micro Keyboard, Quadruped Robot, Light Cube) and instructed club members in 3D modeling, laser cutting, 3D printing, and Arduino.

NYCU Innovation and Entrepreneurship Club *Operating Officer* **Jul. 2023 - Jul. 2024**

- **Coordinated** club operations and events; participated in a student-led startup attempt exploring a tech product.